

Mass is conserved during chemical reactions

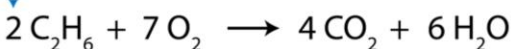
Chemical reactions involve changes in the apportionment of bonding and lone pair electrons.

No matter is created or lost during a chemical reaction. The mass of reactants *must* equal the mass of products.

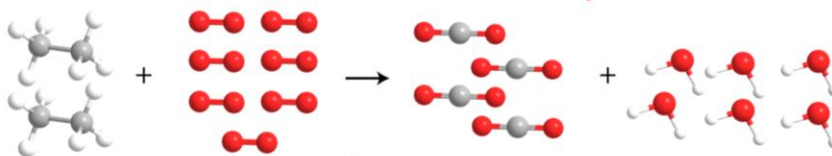
The number of atoms of each element *must* be equal in both the products and reactants.

Example: The complete combustion of ethane

coefficient determines # of molecules



subscript determines molecule identity



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In the upcoming lecture we're going to begin to explore chemical reactions. During a chemical reaction bonds and lone pairs are reapportioned among the substances involved in the reaction. In other words, bonds are broken and formed during reactions, and electrons are rearranged among the structures of the reactants.

Mass is conserved during chemical reactions, meaning that no matter is created or destroyed during the reaction. The identities of all atoms involved remain the same (i.e., the nuclei do not change during a chemical reaction). Therefore, the number of atoms of each element involved in the reaction must be equal in both the products and reactants.

Chemical reactions can be described using a reaction equation like the one presented on the slide. The reaction equation identifies the reactants (on the left side of the arrow) and the products of the reaction (on the right side of the arrow). Each molecular formula in the reaction equation is preceded by a coefficient that identifies the number of moles of each substance that are involved in the reaction. For example, the combustion of ethane involves the reaction of two moles of ethane (C_2H_6) with seven moles of oxygen to produce four moles of carbon dioxide and six moles of water, as shown in the reaction equation on the slide. If no coefficient is shown in front of a molecular formula then the coefficient is assumed to be one. This equation is said to be balanced because the same number of each type of element appears on both sides of the reaction equation. Notice that there are 4 carbon atoms, 12 hydrogen atoms, and 14 oxygen atoms on both the reactant (left) and product (right) sides of the reaction equation.

Use solubility guidelines to predict whether or not a precipitate will form

The following form **soluble** salts

Exceptions

NH_4^+ , Li^+ , Na^+ , K^+ , Rb^+ , and Cs^+	No exceptions
NO_3^- , ClO_4^- , and CH_3COO^-	No exceptions
Cl^- , Br^- , and I^-	Hg_2^{2+} , Ag^+ , Pb^{2+}
SO_4^{2-}	Ca^{2+} , Sr^{2+} , Ba^{2+} , Hg_2^{2+} , Ag^+ , Pb^{2+}

The following form **insoluble** salts

Exceptions

CO_3^{2-} and PO_4^{3-}	NH_4^+ , Li^+ , Na^+ , K^+ , Rb^+ , Cs^+
OH^- and S^{2-}	NH_4^+ , Li^+ , Na^+ , K^+ , Rb^+ , Cs^+ , Ca^{2+} , Sr^{2+} , and Ba^{2+}

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As we will see in class, some reactions are driven by the formation of a solid precipitate from a solution. In order to determine which substances will precipitate we need to be familiar with the solubility trends for common ionic substances. Substances that are soluble in water will not form precipitates, whereas substances that are insoluble will form precipitates.

The top part of the chart on this slide describes substances that are soluble, along with a few exceptions. Ionic compounds in which the cation is NH_4^+ , Li^+ , Na^+ , K^+ , Rb^+ , or Cs^+ will always be soluble, with no exceptions (i.e., the identity of the anion doesn't matter). Similarly, ionic compounds in which the anion is nitrate (NO_3^-), perchlorate (ClO_4^-), or acetate (CH_3COO^-) will always be soluble, with no exceptions (i.e., the identity of the cation doesn't matter). These soluble anions and cations share the common property that they are relatively large and singly charged and, as you would expect from Coulomb's law, only form weak interactions with any counter ion. Ionic compounds in which the anion is chloride (Cl^-), bromide (Br^-), or iodide (I^-) are soluble unless the cation is mercury(I) (Hg_2^{2+}), silver (Ag^+), or lead (Pb^{2+}). For example, aluminum chloride (AlCl_3) would be soluble, but silver chloride (AgCl) would not be soluble. Ionic compounds in which the anion is sulfate (SO_4^{2-}) are usually soluble unless the cation is Ca^{2+} , Sr^{2+} , Ba^{2+} , Hg_2^{2+} , Ag^+ , or Pb^{2+} .

The bottom part of the chart on this slide describes substances that are insoluble, along with a few exceptions. Ionic compounds in which the anion is carbonate (CO_3^{2-}) or phosphate (PO_4^{3-}) are insoluble unless the cation is one of those we listed earlier whose salts are always soluble (NH_4^+ , Li^+ , Na^+ , K^+ , Rb^+ , or Cs^+). Ionic compounds in which the anion is hydroxide (OH^-) or sulfide (S^{2-}) are also insoluble unless the cation is NH_4^+ , Li^+ , Na^+ , K^+ , Rb^+ , Cs^+ , Ca^{2+} , Sr^{2+} , or Ba^{2+} (the "always soluble" cations we listed earlier, plus a couple more).

If two solutions are mixed together that contain an insoluble cation/anion pair, then the pair of ions will react and form an ionic solid that will precipitate from the solution. For example, If a solution of ammonium sulfate is mixed with a solution of barium chloride, the barium ions and sulfate ions will react to form solid barium sulfate. Ammonium chloride, however, is soluble, so the ammonium and chloride ions would not form a precipitate and would instead remained dissolved in the solution as spectator ions.